Ecommerce Customer Churn Analysis

1. Project Objective

The objective of this project is to analyze customer churn patterns in the e-commerce industry using MySQL and identify the factors that influence customer retention. The project focuses on cleaning and transforming customer data, analyzing customer behavior based on tenure, payment preferences, purchase patterns, satisfaction scores, and complaints, and identifying differences between active and churned customers. It also aims to examine customer returns and warehouse-to-home distance to gain a better understanding of customer attrition. The analysis helps generate meaningful insights that can support businesses in improving customer satisfaction, strengthening customer relationships, and developing effective customer retention strategies.

2. Problem Statement

- Analyze customer data to identify patterns and factors contributing to customer churn in the e-commerce industry.
- Identify the number of active and churned customers to understand customer retention levels.
- Examine customer behavior based on tenure, preferred payment methods, order categories, and purchase frequency.
- Evaluate the relationship between customer complaints, satisfaction scores, and churn status.
- Determine the impact of warehouse-to-home distance on customer churn.
- Compare customer preferences based on city tier, gender, marital status, and coupon usage.
- Analyze customer return records and identify customers who have churned and raised complaints.
- Generate actionable insights to support better customer retention strategies and business decisions.

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3. Attribute (Column /Features) Details:

Table 1: customer_churn

S.No	Attribute Name	Data Type	Description
1	CustomerID	INT (Primary Key)	Unique identification number assigned to each customer.

2	Churn	BIT	Indicates whether a customer has left the business (1 = Churned, 0 = Active).
3	Tenure	INT	Number of months the customer has been associated with the company.
4	PreferredLoginDevice	VARCHAR(20)	Device preferred by the customer to log in to the e-commerce application.
5	CityTier	INT	Categorizes the customer's city as Tier 1, Tier 2, or Tier 3.
6	WarehouseToHome	INT	Distance between the customer's home and warehouse, measured in kilometers.
7	PreferredPaymentMode	VARCHAR(20)	Payment method preferred by the customer.
8	Gender	ENUM	Gender of the customer (Male or Female).
9	HourSpendOnApp	INT	Number of hours the customer spends on the e-commerce application.
10	NumberOfDeviceRegistered	INT	Number of devices registered by the customer.
11	PreferedOrderCat	VARCHAR(20)	Product category preferred by the customer when placing orders.
12	SatisfactionScore	INT	Customer satisfaction rating based on their experience.
13	MaritalStatus	VARCHAR(10)	Marital status of the customer (Single, Married, or Divorced).
14	NumberOfAddress	INT	Number of addresses registered by the customer.
15	Complain	BIT	Indicates whether the customer has submitted a complaint (1 = Yes, 0 = No).
16	OrderAmountHikeFromlastYear	INT	Increase in the customer's order amount compared to the previous year.
17	CouponUsed	INT	Number of coupons used by the customer.
18	OrderCount	INT	Total number of orders placed by the customer.
19	DaySinceLastOrder	INT	Number of days since the customer's last order.
20	CashbackAmount	INT	Total cashback amount received by the customer.

Table 2: customer_returns

Attribute Name	Data Type	Description
ReturnID	Int (PrimaryKey)	A unique ID assigned to each product return transaction.
CustomerID	Int	The unique ID of the customer who returned the product.
ReturnDate	date	The date on which the product was returned by the customer.
RefundAmount	Int	The amount of money refunded to the customer for the returned product.

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4. Tools & Technologies

- **MySQL:** DDL, DML Commands, My SQL Constraints, Joins, Aggregate Functions, Subqueries, CASE Expression, NULL Handling, SELECT, WHERE, GROUP BY & HAVING, and ORDER BY & LIMIT

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5. Data Pre-Processing (MySQL)

Tasks Performed:

1. **Database and Table Creation (CREATE):**

Customer_churn table creation

```
5659
5660 •      select * from customer_churn;
5661 •      select CustomerID,OrderAmountHikeFromlastYear,DaySinceLastOrder from customer_churn;
5662
5663
5664                                        /* Data Cleaning:
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2
50009	1	13	Phone	3	9	E wallet	Male	NULL	4	Mobile	3
50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3

tomer_churn 1 | Apply | Revert

Customer returns table creation

```
6035 •    select * from customer_returns;
```

Result Grid | Filter Rows: | Edit:

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990
NULL	NULL	NULL	NULL

customer_returns 44

2. Hndling missing values and outliers:

- **Impute mode for the following columns: Tenure, CoupenUsed, OrderCount**

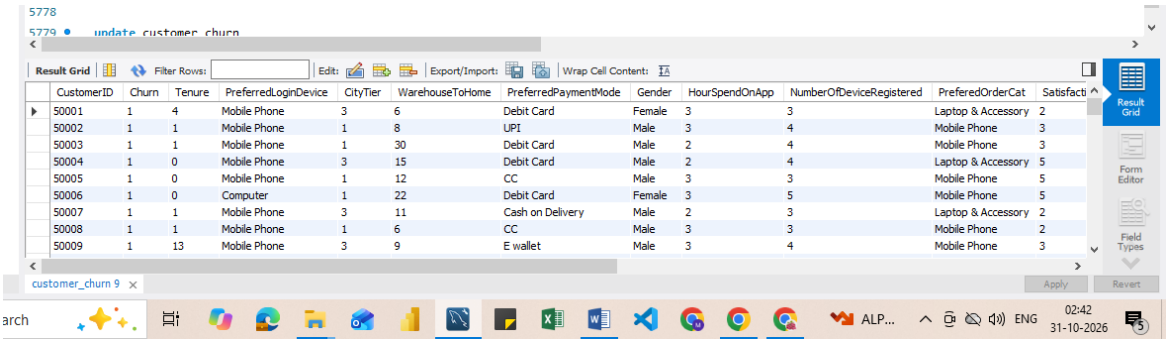
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customerid	tenure	couponused	ordercount
50001	4	1	1
50002	1	0	1
50003	1	0	1
50004	0	0	1
50005	0	1	1
50006	0	4	6
50007	1	0	1
50008	1	2	2
50009	13	0	1
50010	1	1	1

customer_churn 8 | Read Only

3. Dealing with inconsistencies:

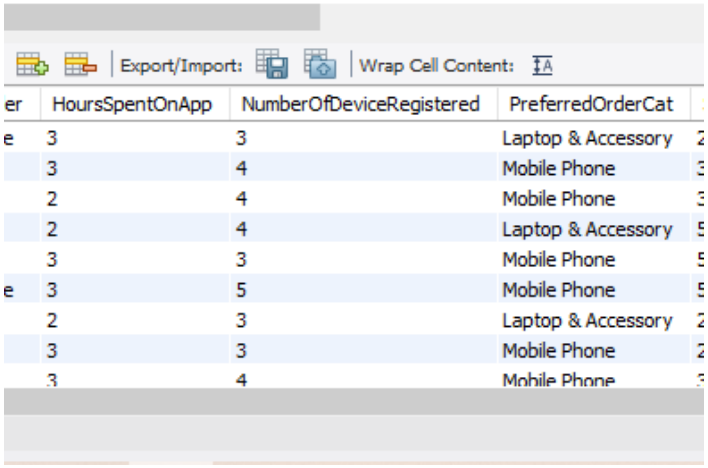
- **Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity**



CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone	3
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone	3
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Mobile Phone	1	12	CC	Male	3	3	Mobile Phone	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	1	Mobile Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	1	Mobile Phone	1	6	CC	Male	3	3	Mobile Phone	2
50009	1	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile Phone	3

4. Data Transformation

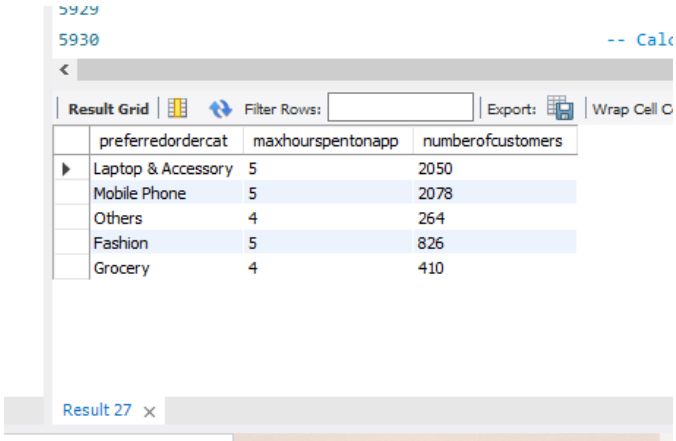
- Rename the column "PreferedOrderCat" to "PreferredOrderCat"
- Rename the column "HourSpendOnApp" to "HoursSpentOnApp"



er	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat
e	3	3	Laptop & Accessory
	3	4	Mobile Phone
	2	4	Mobile Phone
	2	4	Laptop & Accessory
	3	3	Mobile Phone
e	3	5	Mobile Phone
	2	3	Laptop & Accessory
	3	3	Mobile Phone
	3	4	Mobile Phone

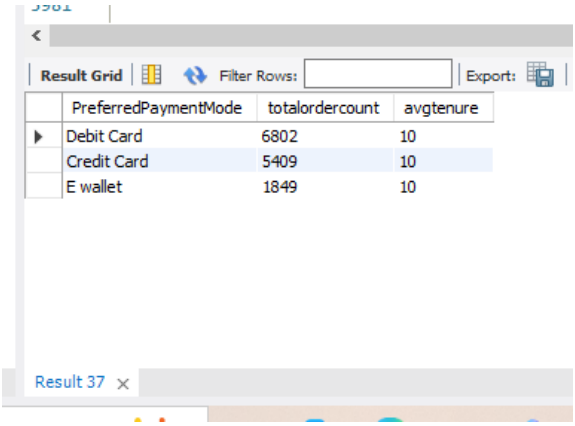
5. Data Exploration and Analysis:

- List the number of customers and the maximum hours spent on the app in each preferred order category



preferredordercat	maxhourspentonapp	numberofcustomers
Laptop & Accessory	5	2050
Mobile Phone	5	2078
Others	4	264
Fashion	5	826
Grocery	4	410

- **Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.**

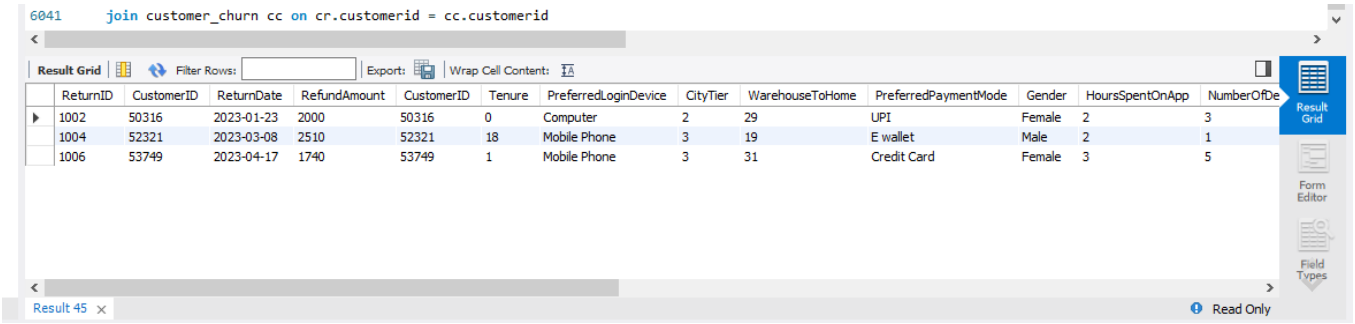


Result Grid | Filter Rows: | Export:

PreferredPaymentMode	totalordercount	avgtenure
Debit Card	6802	10
Credit Card	5409	10
E wallet	1849	10

Result 37

- **Display the return details along with the customer details of those who have churned and have made complaints.**



```
6041    join customer_churn cc on cr.customerid = cc.customerid
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ReturnID	CustomerID	ReturnDate	RefundAmount	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDe
1002	50316	2023-01-23	2000	50316	0	Computer	2	29	UPI	Female	2	3
1004	52321	2023-03-08	2510	52321	18	Mobile Phone	3	19	E wallet	Male	2	1
1006	53749	2023-04-17	1740	53749	1	Mobile Phone	3	31	Credit Card	Female	3	5

Result 45 | Read Only

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6. Key Insights:

The **E-Commerce Customer Churn Analysis** reveals an active base of **4,680 customers against 948 churned users**, with churned accounts heavily concentrated at a low **average tenure of 3 months** and accounting for **₹1,52,030 in sunk cashback costs**. Diagnostic metrics indicate a severe friction point in customer service, as **53.59% of churned users filed complaints**, leaving a poor average satisfaction score of **3 out of 5** among unsatisfied clients. Furthermore, supply chain bottlenecks are evident, with **498 churned customers living in the "Far Distance" category (>15km)** from the warehouse compared to only 1 user in the closest bracket. Predictively, active customers who file complaints or reside over 15km away face an elevated risk of exiting the platform; therefore, prescriptively, the platform must establish a **Priority Ticketing Flag** to fast-track complaint resolutions, transition from upfront cashback to a **Tiered Loyalty Program** that rewards longer tenures, and optimize logistics for distant fulfillment to protect margins and improve retention.

7. Conclusions

This project helped analyze e-commerce customer behavior and understand customer churn patterns using MySQL. Data cleaning, transformation, and SQL queries were used to organize customer information and perform meaningful analysis.

The analysis covered customer retention, purchase behavior, payment preferences, complaints, satisfaction scores, cashback, and product returns.

The findings can help e-commerce businesses identify potential customer retention issues, understand customer preferences, improve customer satisfaction, and develop effective strategies to reduce customer churn.